

Task 3: Why SafeDreamer Violates the Cost Limit

Question

SafeDreamer exceeds the cost threshold ($d = 0$) on multiple SMAC maps. Two candidate causes: (1) world-model error (the model dreams it is safe when it is not), or (2) the Lagrangian multiplier β rises too slowly to enforce the constraint within the training budget.

Ruling out model error (Theorem 1, Task 5)

The Theorem 1 empirical check (Task 5) shows that the world model is accurate enough: the observed gap between real and imagined cost is 50–900 $\times$ smaller than the theoretical bound Δ . Model error is *not* the bottleneck.

β dynamics diagnosis

We pulled the Lagrangian multiplier (**Agent/Lagrangian**) and episode cost (**main/cost**) timeseries from WandB for 5 SMAC maps at $d = 0$ (3 seeds each). Figures 1 and 2 show the results.

Key finding: β never converges. On all 5 maps, β rises linearly throughout training and is still climbing at 100k steps. There is no plateau, no oscillation — the gradient-ascent update (lag-lr = 10^{-5}) is simply too slow.

Map	β at 100k	Cost at 100k	Initial cost	Verdict
MMM	0.50	$\sim 7\text{--}8$	~ 10	worst — β rising but cost barely moves
1c3s5z	0.30	$\sim 4\text{--}5$	~ 9	bad — cost dropping but far from 0
8m	0.50	~ 3	~ 8	medium — cost dropping, β still rising
3m	0.13	~ 0.5	~ 3	mild — almost working, needs more time
2m_vs_1z	0.025	~ 0.2	~ 2	best — low cost, low β needed

Table 1: β and cost at end of training (100k steps, $d = 0$, 3-seed mean).

Severity scales with map difficulty. On easy maps (2m_vs_1z, 3m), the initial cost is low so even a small β suffices. On hard maps (MMM, 1c3s5z, 8m), the initial cost is 8–10 dead allies per episode; β at 0.3–0.5 is not large enough to outweigh the reward signal, so the policy keeps violating.

Conclusion

The cost-above-threshold problem is a **Lagrangian convergence** problem, not a model-accuracy problem:

- The multiplier update is too slow relative to the 100k training budget.
- Given more training time, β would eventually rise enough (the trend is monotonic), but SafeDreamer’s sample efficiency advantage would be lost.
- A faster multiplier mechanism — specifically PID-Lagrangian (Task 4) — should fix this by adding proportional and derivative terms that respond to violation magnitude and trend, not just accumulated error.

Figures

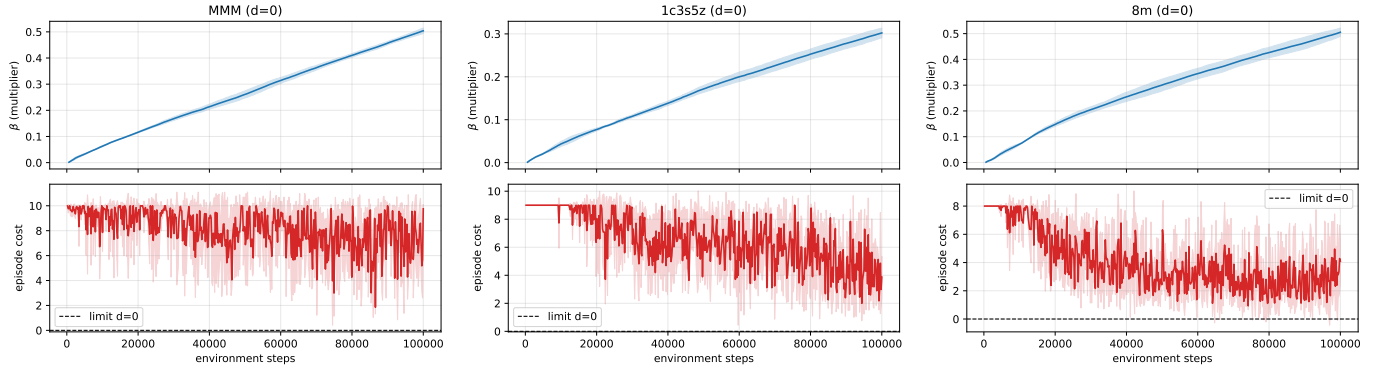


Figure 1: β dynamics on violating maps (MMM, 1c3s5z, 8m). β rises linearly; cost remains well above $d = 0$ at 100k steps.

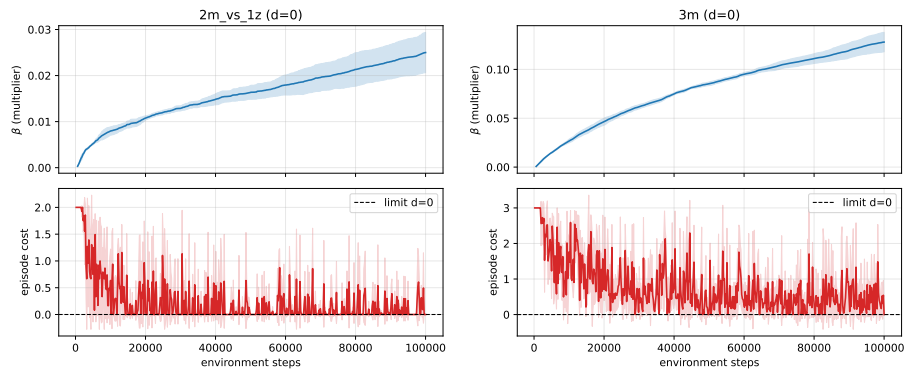


Figure 2: β dynamics on clean maps (2m_vs_1z, 3m). Lower initial cost means a small β is sufficient; cost approaches 0 by 100k.